

PROGRAMMING WITH C++

Course Code:	CS3651-1	Course Type	AEC
Teaching Hours/Week (L: T: P)	1:0:2	Credits	02
Total Teaching Hours	15+0+26	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering

Course Objectives:

1.	Understand the basic principles of Object-oriented programming and, enables implementing through C++ programming language.
2.	Outline the structure of C++ program, use of its operators, expressions and the concepts of classes and objects to write a program
3.	Design the program using the concepts of constructors, different types of functions.
4.	Apply the concepts of inheritance, overloading for programming.
5.	Demonstrate the use virtual functions, templates, exception handling and Operator overloading.

Principles of object – oriented programming:

A look at Procedure Oriented Programming, object Oriented Programming Paradigm, Basic Concepts of OOP, Benefits of OOP, Applications of OOP.

Beginning with C++:

What is C++, Structure of C++ program, Basic Data types, derived data types, user defined data types, dynamic initialization of variables, reference variables, scope resolution operator, memory management operators, type cast operators.

Functions in C++:

Function prototyping, Inline Functions, Default Arguments, Function Overloading

Classes and objects:

Specifying a Class, Defining Member Functions, Static Data Members, and Static Member Functions. Arrays of Objects, Objects as Functions Arguments, Friend Functions, Returning Objects.

Constructors and Destructors

Introduction, Constructors, Parameterized Constructors, Multiple Constructors in a Class. Constructors with Default Arguments, Copy Constructors, Dynamic Constructors, Destructors.

Inheritance: Introduction, Defining Derived Classes, Single Inheritance, Protected Access Specifier, Multilevel Inheritance, Multiple Inheritance, Hierarchical Inheritance, Hybrid Inheritance, Virtual Base Classes, Abstract Classes, Constructors in Derived Classes.

Op Operator Overloading and Type Conversions: Defining Operator Overloading, Overloading Increment and Decrement Operators (Prefix and Postfix), Rules for overloading operators.

Exception Handling: Basics of Exception Handling, Exception Handling Mechanism.

Pointers, Virtual Functions: Pointers to Objects, Pointers to Derived Classes, Virtual Functions, Pure Virtual Functions.

List of Experiments

PART-A

1	Write C++ program to design a class called BankAccount. Include following data members like name of the depositor, account number and balance. Use following member functions a) to initialize values b) deposit an amount c) to withdraw an amount d) to display name and balance.
2	Simple class and objects creation. Write a C++ program to create a class called COMPLEX and implement the following overloading function ADD that return a COMPLEX number. i. ADD(a,c2);- where a is an integer(real part) & c2 is a complex no.

	ii. ADD(c1,c2);- where c1 & c2 are complex nos Use Function overloading(ADD) and Friend function concept for the implementation.
3	Write a C++ program with class Time with data members that represents hours and minutes. Include appropriate member functions to compute time in hours and minutes . (Use of objects as arguments).
4	Given that an EMPLOYEE class contains following members. Data members:Eno, Ename and salary Member functions: to read the data , to print data members.Write a C++ program to read the data of N employees and display details of each employee.(use Array of objects concept).
5	Write a C++ program to demonstrate the uses of constructors in derived class concept. (Any inheritance you can use but constructors in base class should have at least one parameter.)

PART-B

6	Write a C++ program to create a class sample with integer ,character and float data members. Demonstrate Constructor Overloading on this class with all types of constructors including default argument constructor.
7	Write a C++ program to demonstrate the working of dynamic constructors using a class STRNG with string as data member inside the class, include appropriate member functions to display the object data and a member function to concatenate two strings.
8	Write a C++ program for the diagram using Hierarchical inheritance. Use your own data members and member functions to display student details.
9	Create a base class shape to store two double type values. Derive two specific classes called triangle and rectangle from the base shape. Add to the base class, a member function get_data() to initialize base class data members and another function display_area() to compute and display the area of figures. Make display_area() as a virtual function and redefine this function in the derived classes to suit the requirements.(Area of rectangle= $x*y$, Area of triangle= $1/2*x*y$).
10	Write a C++ program to overload binary + and – operator to add and subtract two complex numbers. Define relevant data members and member functions for reading and displaying the complex objects

Course Outcomes: At the end of the course student will be able to

1.	Apply the basic concepts of C++ for problem solving and build programs using basic constructs of C++.
2.	Develop solutions to problems using the concepts of functions and its variants, classes and objects.
3.	Apply the concepts of constructors and its variants to demonstrate the object initialization.
4.	Develop programs in C++ that demonstrate code re-usability and polymorphism using the concept of inheritance and operator overloading and utilize concept of exception handling, pointers, an and virtual functions.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3651-1.1	2	2	3	-	2	-	-	1	1	1	-	1	1	2	3
CS3651-1.2	2	2	3	-	2	-	-	1	1	1	-	1	1	2	3
CS3651-1.3	2	2	3	-	2	-	-	1	1	1	-	1	1	2	3
CS3651-1.4	2	2	3	-	2	-	-	1	1	1	-	1	1	2	3

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	E.Balagurusamy, Object – Oriented Programming with C++, Eight Edition, Tata McGraw Hill. 2018.
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REFERENCE BOOKS:

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| 1. | Robert Lafore: Object – Oriented Programming in Turbo C++, 4th Edition , 2008. |
| 2. | Herbert Schildt: C++ The Complete Reference, 4th Edition, 2017. |

E Books / Online Resources

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| 1. | Tour of C++ , Second edition, Bjarne Stroustrup, |
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MOOCs

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| 1. | https://www.udemy.com/learn-advanced-c-programming |
| 2. | https://nptel.ac.in/courses/106105151 |